Emergent Time as Information-Geometric Path Length in Open Quantum Dynamics

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Abstract. We develop a dynamical foundation for the information-geometric framework of emergent time introduced in a companion paper. Starting from a closed tripartite system *SOE* (system–observer–environment) evolving unitarily, we show that the binary anomaly variable σn is generated dynamically by the reduced evolution of the observer, not postulated. We define the reference state $\rho G^{(R)}$ as the projection of the complete description onto the observer-accessible algebra, providing a precise quantum bridge between the abstract complete description G and a computable density matrix. The emergent temporal parameter τ is identified with the accumulated Bures distance traversed by the observer's reduced state, i.e., the information-geometric path length of the partial description. We compute this functional explicitly for a qubit undergoing pure dephasing, obtaining closed-form expressions for F(t), Δ(t), and τ(t). The example demonstrates that temporal orientation, monotonicity, and finiteness of accumulated time all follow from standard open-system dynamics without additional assumptions. The results strengthen the claim that time is not a fundamental coordinate but the length of the trajectory traced by a partial description.

1. Introduction

In the companion paper [1], we developed an information-geometric framework in which a complete description G of a physical system is atemporal, while a temporal parameter τ emerges for observers with partial access to G. The binary anomaly variable $\sigma n \in \{-1, +1\}$ tracked whether successive updates of the partial description moved toward or away from G, and τ was defined as the accumulated signed information distance.

That framework left three foundational questions open. First, the reference state ρG was described as the "complete description G" without specifying its quantum representative for a given observer. Second, the variable σn was introduced as a formal quantity without demonstrating that it arises from standard quantum dynamics. Third, the limit from discrete updates to a continuous temporal parameter lacked a physical justification tied to an underlying Hamiltonian evolution.

This paper resolves all three. Our contributions are:

(i) A precise definition of $\rho^{G(R)}$ as the projection of G onto the observer-accessible algebra (Section 2), providing the quantum bridge from abstract complete description to computable density matrix.

(ii) A proof that σn is generated dynamically by the reduced evolution $\rho^{R(n)} = \mathrm{Tr}_{SE}[\rho SOE^{(n)}]$ under unitary global dynamics (Section 3), so it is not a postulate but a consequence of ordinary quantum mechanics.

(iii) The identification of τ with the information-geometric path length of the observer's reduced state trajectory (Sections 4–5), and a complete closed-form computation for a qubit undergoing pure dephasing (Section 6).

The central result is:

$$\tau(t) = \int_0^t |d/ds\, D_B(\rho_G, \rho_O(s))|\, ds,$$

which identifies emergent time with the total variation of the Bures distance along the observer's trajectory. This is not a definition imposed from outside the framework; it follows from the discrete definition of τ in [1] when the reduced state traces a differentiable curve in the space of density matrices.

The paper is organized as follows. Section 2 defines the observer-projected reference state. Section 3 derives the dynamical origin of σ^n. Section 4 defines τ as a discrete path length. Section 5 passes to the continuous limit. Section 6 works out the qubit dephasing example in full. Section 7 discusses implications.

2. From G to the Observer-Projected Reference State

The complete description G of [1] is a maximal statistical manifold satisfying conditions C1–C3. For a quantum system, G corresponds to the full space of density matrices $D(\mathfrak{H})$ on a Hilbert space $\mathfrak{H}$. A partial observer O has access only to a subalgebra $A^R \subset \mathcal{B}(\mathfrak{H})$, determined by which degrees of freedom survive the partial trace over system S and environment E:

$$\rho_O^{(n)} = Tr_{SE}[\rho_{SOE}^{(n)}].$$

The quantum representative of G as seen from the observer is not G itself, but its projection onto A^R. We define:

Definition 1 (Observer-Projected Reference State). The reference state $\rho^{G(R)}$ is

$$\rho_G^{(R)} := \operatorname{argmin}_{\rho \in A_O} D_B(\rho, \rho_O),$$

where D_B is the Bures distance and A^R is the set of states compatible with the projection of G onto the observer-accessible degrees of freedom.

Operationally, $\rho^{G(R)}$ is the state within the observer's accessible algebra that is closest to the observer's current state ρ^R. It is the quantum representative of the complete model as seen from the observer's reduced perspective. Crucially, it is well-defined without reference to the full G: it is determined by the observer's algebra A^R and the ambient metric.

In the qubit dephasing example of Section 6, ρ^G is the fully dephased state $\rho^G = \mathrm{diag}(p, 1-p)$, which is the unique state in the pointer basis algebra that is closest (in Bures distance) to the coherent initial state $\rho^R(0)$. This choice is both physically transparent and computationally tractable.

3. Dynamical Origin of the Anomaly Variable

Consider a closed tripartite system SOE (system S, observer O, environment E) evolving unitarily:

$$\rho_{SOE}^{(n+1)} = U_n\, \rho_{SOE}^{(n)}\, U_n^\dagger.$$

The observer has access only to the reduced state:

$$\rho_O^{(n)} = Tr_{SE}[\rho_{SOE}^{(n)}].$$

Define the information distance between the reference state and the observer's reduced state:

$$\Delta_n = D_B(\rho_G, \rho_O^{(n)}),$$

where $D_B(\rho, \sigma) = \sqrt{2-2\sqrt{F(\rho,\sigma)}}$ and $F(\rho,\sigma) = \mathrm{Tr}(\sqrt{\sqrt{\rho}\,\sigma\,\sqrt{\rho}})$ is the Uhlmann fidelity.

Theorem 4 (Dynamical Origin of the Anomaly Variable). Let the closed system SOE evolve unitarily. Let $\rho^{R(n)} = \mathrm{Tr}_{SE}[\rho_{SOE}^{(n)}]$ be the observer's reduced state and $\Delta_n = D_B(\rho^G, \rho^{R(n)})$ the corresponding Bures distance. Then the binary variable

$$\sigma_n := \mathrm{sgn}(\Delta_n - \Delta_{n+1})$$

is generated dynamically by the reduced evolution. Specifically: $\sigma_n = +1$ when the observer's reduced state moves closer to ρ^G (information gain), and $\sigma_n = -1$ when it moves farther (information loss or decoherence fluctuation).

Proof. The sequence $\rho^{R(n)}$ is fully determined by the unitary U_n and the initial state $\rho_{SOE}^{(0)}$ via the partial trace. Hence Δ_n is a computable function of the global dynamics, and $\sigma_n = \mathrm{sgn}(\Delta_n - \Delta_{n+1})$ is determined at each step by the reduced evolution alone. It is not a free parameter. □

This theorem closes the first gap in [1]: the anomaly variable is not postulated but derived from the same unitary dynamics that governs the rest of quantum mechanics. Irreversibility enters only through the partial trace, not through the global dynamics.

4. Emergent Time as Discrete Information-Geometric Path Length

We now reinterpret the accumulated temporal parameter τ of [1] as the total variation of the Bures distance along the observer's trajectory. In the discrete setting, define:

$$\tau_N = \sum_{n=0}^{N-1} |\Delta_{n+1} - \Delta_n|.$$

This is the total variation of the sequence (Δ_n), i.e., the length of the trajectory traced by $\rho^{R(n)}$ in Bures space, measured in units of informational distance from ρ^G.

The connection to the definition of τ in [1] is direct: since $\sigma_n = \mathrm{sgn}(\Delta_n - \Delta_{n+1})$, we have $|\Delta_{n+1} - \Delta_n| = \sigma_n(\Delta_n - \Delta_{n+1})$, so

$$\tau_N = \sum_{n=0}^{N-1} \sigma_n (\Delta_n - \Delta_{n+1}),$$

which coincides with the temporal parameter of [1] when $\sigma_n \geq 0$ dominates (monotone regime). The path-length interpretation, however, is more general: it applies even when Δ_n is non-monotone, because the absolute value removes sign ambiguity.

Remark. The path-length definition τ_N is always non-negative and non-decreasing by construction. It cannot decrease as the observer accumulates updates. This is the information-geometric analogue of the irreversibility of thermodynamic entropy increase.

5. Continuous Limit

Suppose the discrete steps $n \to n+1$ correspond to a physical time increment $\varepsilon \to 0$, and that the map $n \mapsto \rho^{R(n)}$ converges to a differentiable curve $\rho^R(t)$ in the space of density matrices. Then:

$$\tau(t) = \int_0^t |d/ds\, D_B(\rho_G, \rho_O(s))|\, ds.$$

This is the arc length of the curve $s \mapsto D_B(\rho^G, \rho^R(s))$ in $\mathbb{R}_{\geq 0}$, not the arc length of the curve $\rho^R(s)$ in the Bures manifold itself. The distinction matters: τ measures how much the informational gap changes, not how much the state moves in total.

The continuous limit is justified when the underlying Hamiltonian evolution is smooth and the environmental coupling generates a Lindblad-type master equation for $\rho^R(t)$. In the qubit dephasing example of Section 6, the master equation is

$$d\rho_O/dt = \gamma(\sigma_z \rho_O \sigma_z - \rho_O),$$

which admits an exact analytical solution and makes $\tau(t)$ computable in closed form.

Definition 2. The emergent time parameter for observer O is $\tau(t)$ as defined above. It is a functional of the reduced trajectory, not an external coordinate. The direction of τ is determined by the sign of $d\Delta/dt$: $\sigma(t) = \mathrm{sgn}(-d\Delta/dt)$.

6. Qubit Dephasing: Closed-Form Computation

6.1 Setup

Let the observer's initial state and reference state be the qubit matrices

$$\rho_O(0) = \begin{pmatrix} p & c \\ c^* & 1-p \end{pmatrix}, \quad \rho_G = \begin{pmatrix} p & 0 \\ 0 & 1-p \end{pmatrix},$$

where $p \in (0,1)$ and $|c|^2 \le p(1-p)$ (positivity constraint). Under pure dephasing with rate $\gamma > 0$,

$$\rho_O(t) = \begin{pmatrix} p & c\, e^{-\gamma t} \\ c^*\, e^{-\gamma t} & 1-p \end{pmatrix}.$$

The reference state ρ^G is the fully dephased pointer state: the unique state in the diagonal (pointer) algebra closest to $\rho^R(0)$ in Bures distance. As $t \to \infty$, $\rho^R(t) \to \rho^G$ monotonically.

For a pure state $|+\rangle\langle +|$ ($p = 1/2$, $|c|^2 = 1/4$, i.e., $u = 1$ with $u = |c|^2/[p(1-p)]$), the fidelity between the initial pure state and $\rho^G = I/2$ is:

$$F(|+\rangle\langle +|, I/2) = \langle +|\, (I/2)\, |+\rangle = 1/2,$$

which gives $\Delta(0) = \sqrt{2-2\sqrt{1/2}} = \sqrt{2-\sqrt{2}} \approx 0.765$.

6.2 Fidelity and Bures Distance

The Uhlmann fidelity between $\rho^G = \mathrm{diag}(p, 1-p)$ and $\rho^R(t)$ is computed via $F = \mathrm{Tr}(M) + 2\sqrt{\det M}$, where $M = \sqrt{\rho^G}\, \rho^R(t)\, \sqrt{\rho^G}$. Evaluating explicitly:

$$F(t) = 1 - 2p(1-p) \cdot [1 - \sqrt{1 - u\, e^{-2\gamma t}}],$$

where $u = |c|^2/[p(1-p)] \in [0,1]$ is the normalized initial coherence. Its time derivative is

$$dF/dt = 2\gamma p(1-p)\, u\, e^{-2\gamma t} / \sqrt{1 - u\, e^{-2\gamma t}} > 0,$$

which is strictly positive for all $t > 0$. Therefore $F(t)$ increases monotonically, $\sqrt{F(t)}$ increases, and

$$\Delta(t) = \sqrt{2 - 2\sqrt{F(t)}}$$

decreases strictly. This establishes:

Proposition 1. Under pure dephasing, $\sigma(t) = \mathrm{sgn}(-d\Delta/dt) = +1$ for all $t > 0$. The observer consistently gains information relative to ρ^G.

6.3 Emergent Time

Since $\Delta(t)$ decreases monotonically, the path-length integral collapses to

$$\tau(t) = \int_0^t |d\Delta/ds|\, ds = \Delta(0) - \Delta(t).$$

$\tau(t)$ is monotone, non-negative, and saturates at

$$\tau(\infty) = \Delta(0) < \infty.$$

For the canonical case ($p = 1/2$, $u = 1$, $\gamma = 1$):

$$\tau(\infty) = \sqrt{(2 - \sqrt{2})} \approx 0.765.$$

This is finite, exact, and independent of any free parameter once the initial state and reference state are fixed.

Table 1. Numerical behavior of $\tau(t)$ for the canonical dephasing example with $\rho^R(0) = |+\rangle\langle +|$, $\rho^G = I/2$, and $\gamma = 1$. Here $\tau(t) = \Delta(0) - \Delta(t)$ and $\Delta(0) = \sqrt{(2-\sqrt{2})} \approx 0.765$.

t	e^{-2t}	F(t)	Δ(t)	τ(t)	τ(t)/τ(∞)
0	1.000	0.500	0.765	0.000	0.000
0.5	0.368	0.898	0.324	0.441	0.576
1.0	0.135	0.965	0.188	0.577	0.754
1.5	0.050	0.987	0.112	0.653	0.853
2.0	0.018	0.995	0.068	0.698	0.911
3.0	0.002	0.999	0.025	0.740	0.967
∞	0.000	1.000	0.000	0.765	1.000

Approximately 85% of the total temporal accumulation occurs for $t \leq 1.5$, with exponential saturation thereafter.

7. Implications

7.1 Dynamical Origin of Temporal Orientation

The anomaly variable $\sigma(t)$ is not an independent input but is generated dynamically by the reduced evolution of the observer. In the dephasing model, monotonic decay of coherence implies

$$\sigma(t) = +1 \quad \forall t > 0,$$

showing that a consistent temporal orientation emerges from standard open-system dynamics without additional assumptions. The direction of time is a consequence of the contractivity of the dephasing channel, not a postulate.

7.2 Emergent Time as Information-Geometric Length

The temporal parameter is exactly

$$\tau(t) = \Delta(0) - \Delta(t),$$

which identifies time with the accumulated information-geometric change of the observer's state. This provides a concrete realization of the idea that time is not fundamental but derived from the trajectory of partial descriptions. The formula has a clear operational meaning: τ measures how

much of the total informational gap between the initial coherent state and the equilibrium state has been closed by decoherence.

7.3 Finiteness of Temporal Length

The saturation $\tau(\infty) = \Delta(0)$ shows that the total amount of temporal evolution accessible to an observer is finite. The total "measurable time" is bounded by the initial informational mismatch between the observer's state and the coarse-grained equilibrium. An observer that begins closer to equilibrium has less temporal length to accumulate. This is consistent with the interpretation of $\Delta(0)$ as a measure of informational distance from stasis.

7.4 Arrow of Time and Decoherence

The emergence of a preferred temporal direction is directly tied to decoherence. The monotonic approach of $\rho^R(t)$ to ρ^G ensures that $d\Delta/dt < 0$, and therefore $\tau(t)$ is monotone. The arrow of time is identified with the irreversible loss of phase information under the dephasing channel. This is not a thermodynamic assumption; it follows from the positivity of dF/dt proved in Section 6.2.

7.5 Universality and CPTP Contractivity

The result does not depend on the specific form of the Hamiltonian, but only on two structural features: (a) the dephasing channel is a CPTP map, and (b) the Bures distance is contractive under CPTP maps ($DB(\Lambda(\rho), \Lambda(\sigma)) \leq DB(\rho, \sigma)$ for all CPTP maps Λ). Any CPTP map that sends $\rho^R(t)$ monotonically toward ρ^G will produce the same qualitative behavior: $\sigma(t) = +1$ throughout, τ monotone, $\tau(\infty)$ finite. This suggests that the emergence of time as information-geometric length is a universal feature of open quantum systems approaching equilibrium, not a special property of dephasing.

Taken together, these results show that the abstract framework of [1] admits a fully explicit realization in a minimal physical model. The emergence of time, the directionality of temporal flow, and the finiteness of temporal accumulation all follow from standard quantum dynamics combined with information-geometric structure.

8. Relation to Prior Work

The identification of time with information-geometric distance has appeared in various forms. Wootters [2] proposed using statistical distance as a measure of distinguishability between states; Page and Wootters [3] constructed relational time from correlations in an entangled bipartite system. The present framework differs in two respects: τ is a path length (total variation), not a distance between two fixed states, and it is defined for a tripartite SOE system rather than a bipartite one.

The contractivity of the Bures distance under CPTP maps was established by Uhlmann [4] and plays a central role here. The use of the Bures metric specifically, rather than other elements of the Petz family, follows from its status as the minimal monotone metric and its continuity extension to pure states [5].

The structural parallel with the Wheeler–DeWitt equation, noted in [1], is preserved here: the complete global evolution $\rho SOE^{(n)}$ is unitary and contains no preferred time, while τ emerges only after restriction to the observer's reduced state. This is consistent with the Page–Wootters mechanism [3] and with relational quantum mechanics [6], but the mechanism of emergence (path length of the Bures distance) is distinct.

9. Connection to CMB Observations

9.1 From Primordial Modes to aℓm

The empirical connection to the CMB requires a non-arbitrary bridge between the information-geometric framework and the observed angular coefficients. The appropriate bridge is not the TT power spectrum alone, but the reduced quantum state of primordial perturbation modes.

Each primordial scalar mode k may be modeled as a quantum harmonic oscillator. During inflation, such modes are driven into highly squeezed states. After horizon crossing, the observable sector is described by the reduced state

$$\rho_O^{(k)} = Tr_E[\rho^{(k)}_{OE}],$$

where the trace is over inaccessible environmental degrees of freedom: unobserved phases, superhorizon correlations, and coarse-grained gravitational variables. For a squeezed mode with squeezing parameter r_k, tracing over the companion mode $k' = -k$ yields a thermal state with mean occupation $n_k = \sinh^2(r_k)$. Truncated to the two lowest levels (valid for well sub-horizon modes with $n_k \ll 1$), this gives

$$\rho_O^{(k)} \approx diag(1/(1+n_k), n_k/(1+n_k)), \quad \rho_G^{(k)} \approx diag(1/(1+n_k^{\Lambda CDM}), n_k^{\Lambda CDM}/(1+n_k^{\Lambda CDM})).$$

The information-geometric temporal length of mode k is then

$$\tau_k = \int |d/ds\, D_B(\rho_G^{(k)}, \rho_O^{(k)}(s))|\, ds.$$

This structure is projected onto multipoles via the standard CMB transfer function $T_\ell(k)$:

$$a_{\ell m} = \int d^3k\, T_\ell(k)\, \zeta_k\, Y^*_{\ell m}(\hat{k}), \quad \tau_{\ell m} \sim \int d^3k\, |T_\ell(k)|^2\, |Y_{\ell m}(\hat{k})|^2\, \tau_k.$$

9.2 Why the TT Power Spectrum Alone Is Insufficient

The TT angular power spectrum averages over azimuthal indices m:

$$C_\ell = (1/(2\ell+1)) \sum_m |a_{\ell m}|^2,$$

discarding phase information and all angular covariance structure between distinct modes (ℓ, m) and (ℓ', m'). A direct calculation shows that, to first order in the standardized TT residual $z_\ell = (D_\ell^{obs} - D_\ell^{th})/\sigma_\ell$, the Bures distance between the observed and predicted mode states reduces to

$$\Delta_\ell = D_B(\rho_G^{(\ell)}, \rho_O^{(\ell)}) \approx C\, |z_\ell|, \quad C = \sqrt{n_k} \cdot \sigma_{rel} / (2(1+n_k)),$$

where σ_{rel} is the relative spectral uncertainty. Consequently $\tau_\ell \propto |z_\ell|$, and any community structure obtained from τ_ℓ in a TT-only analysis is equivalent, up to rescaling, to the structure obtained directly from $|z_\ell|$. This makes the TT-only result circular: it recovers what was already used to build the graph.

We therefore regard the TT-only analysis of [7] as a consistency check, not as an independent empirical test of the information-geometric framework.

9.3 Non-Trivial Prediction for Full aℓm Maps

The prediction becomes genuinely non-trivial at the level of the full angular coefficients aℓm. The key quantity is not the information-geometric length τ of individual modes, but the *informational correlation between pairs of modes*. Define the Bures mutual information between modes i = (ℓ, m) and j = (ℓ', m'):

$$I_{ij} = D_B(\rho_O^{\{ij\}}, \rho_O^i \otimes \rho_O^j),$$

the Bures distance between the joint two-mode state and the tensor product of the individual reduced states. This quantity vanishes if and only if the two modes are informationally independent. It is non-zero whenever the joint state contains correlations not captured by the individual amplitudes.

Two graphs can now be constructed on the same node set {(ℓ, m)}:

Power graph: $W^{(P)}_{ij} = \mathrm{corr}(|a_i|^2, |a_j|^2)$, capturing correlations of squared amplitudes (second-moment structure).

Information-geometric graph: $W^{(\tau)}_{ij} = I_{ij}$, capturing Bures mutual information between pairs of modes (beyond second-moment structure).

For ideal full-sky Gaussian statistically isotropic ΛCDM realizations, off-diagonal informational correlations vanish in expectation. In realistic Planck-like maps, however, mask, beam, anisotropic noise, lensing, and component separation induce a non-zero null distribution for $W^{(\tau)}$. The prediction is therefore not that $W^{(\tau)}$ vanishes for ΛCDM, but that the observed community mismatch lies significantly outside the Planck-like null distribution. The framework therefore predicts:

$$\mathrm{Comm}(W^{(\tau)}) \neq \mathrm{Comm}(W^{(P)}) \quad \text{[Planck PR4 data]},$$

$$\mathrm{Comm}(W^{(\tau)}) \approx \mathrm{Comm}(W^{(P)}) \quad \text{[}\Lambda\text{CDM simulations]}.$$

The mismatch is quantified by the normalized variation of information (NVI) between the two community partitions:

$$\mathrm{NVI}(\mathrm{Comm}(W^{(P)}), \mathrm{Comm}(W^{(\tau)})),$$

compared against an ensemble of ΛCDM simulations processed through the same mask, beam, noise, and component-separation pipeline as the Planck PR4 maps.

9.4 Falsification Criterion and Connection to Non-Gaussianity

The falsification criterion is explicit. If the observed NVI lies within the ΛCDM simulation distribution, the information-geometric prediction is not supported by the data. If it lies significantly outside that distribution (> 2σ), the framework identifies a phase-sensitive correlation structure in the CMB not reducible to TT power anomalies.

The connection to primordial non-Gaussianity is structural: the Bures mutual information I_{ij} between modes i and j receives contributions from all moments of the joint distribution, not only the second. For Gaussian fields, $I_{ij} = 0$ by definition. For local-type non-Gaussianity, the bispectrum is expected to induce non-zero informational correlations between modes; the leading scaling may be quadratic in the non-Gaussian amplitude for distance-based measures, but the precise dependence requires explicit computation and is left as an open problem. The framework therefore predicts that $W^{(\tau)}$ is sensitive to primordial non-Gaussianity in a manner independent of and complementary to standard f_{NL} estimators, which are optimized for the bispectrum directly.

Three predictions follow in order of increasing strength. First, for ΛCDM-consistent Planck PR4 simulations, $\mathrm{NVI}(\mathrm{Comm}(W^{(P)}), \mathrm{Comm}(W^{(\tau)}))$ should be small and consistent with the null distribution. Second, if the observed CMB contains the TT anomalies already detected in [7], the full $a_{\ell m}$ maps should show NVI outside the simulation distribution. Third, and most ambitiously, the community structure of $W^{(\tau)}$ might exhibit a factorization compatible with the proposed 1+3 emergent structure — the same 2×2 pattern (sign × regime) detected in the TT analysis, but extended to three spatial dimensions via the full angular information. This last possibility should be treated as a high-risk prediction rather than an established consequence of the present

paper: it requires completing the analysis of the $a_{\ell m}$ maps and verifying the dimensional structure independently.

Epistemic status: the bridge from primordial squeezed states to observable $a_{\ell m}$ involves the approximation of truncating each mode to a two-level system, the identification of the environmental trace with superhorizon decoherence, and the use of the transfer function $T_\ell(k)$ as a classical projection. These approximations are standard in inflationary cosmology but introduce model dependence. The predictions above should therefore be treated as working hypotheses pending explicit computation of I_{ij} from Planck PR4 data.

10. Open Problems

Non-monotone trajectories. The dephasing example produces a monotone $\Delta(t)$ and thus $\tau = \Delta(0) - \Delta(t)$. For more general CPTP maps, $\Delta(t)$ may oscillate before settling. In that case τ is the total variation, strictly larger than $\Delta(0) - \Delta(\infty)$. The physical interpretation of τ in oscillatory regimes—and whether the oscillations have observable consequences—remains open.

Derivation of ρ^G from G. Definition 1 characterizes $\rho^{G(R)}$ as the Bures projection of G onto the observer's algebra. A complete derivation requires specifying G as a density matrix on the full Hilbert space and performing the partial trace explicitly. For interacting many-body systems, the resulting $\rho^{G(R)}$ may depend on the interaction Hamiltonian in a non-trivial way. Characterizing this dependence is an open problem.

Quantitative predictions. The dephasing rate γ appears as a free parameter in the present example. A complete physical theory should derive γ from the system–environment coupling. Whether the information-geometric framework constrains γ, or predicts relations between γ and other observables (e.g., the decoherence rate prediction of [1, Section 7]), requires further work.

Extension to a_{lm} analysis. The companion empirical paper [7] verified the temporal-dimension prediction of the Xarxa Atemporal framework using the CMB power spectrum (one-dimensional in angular scale l). A full verification of the Minkowski (1,3) signature prediction requires extending the analysis to the angular coefficients a_{lm} of the Planck PR4 maps. The identification of τ with Bures path length suggests a specific form for the temporal component of the emergent metric in that analysis.

11. Conclusions

We have shown that the three foundational gaps in the information-geometric framework of [1] can be closed within standard quantum mechanics:

(1) **The reference state $\rho^{G(R)}$** is the Bures projection of the complete description onto the observer-accessible algebra. It is computable from the partial trace structure alone.

(2) **The anomaly variable σ_n** is generated dynamically by the reduced evolution $\rho^{R(n)} = \mathrm{Tr}_{SE}[\rho_{SOE}^{(n)}]$. It is not a postulate.

(3) **The emergent time τ** is the information-geometric path length of the observer's reduced trajectory in Bures space, computable in closed form for the qubit dephasing model:

$$\tau(t) = \Delta(0) - \Delta(t), \quad \tau(\infty) = \sqrt{2-\sqrt{2}} \approx 0.765.$$

The example demonstrates monotonicity, directedness, and finiteness of accumulated time as consequences of decoherence, without any additional assumptions. The arrow of time is the contractivity of the dephasing channel applied to Bures distance.

The central claim of the framework is confirmed in a minimal computable model: *time is not a fundamental coordinate but the length of the trajectory traced by a partial description.*

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